

Qatar-9b

R-2402

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250320 · created 2026-08-21T00:34:09+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460754.6393 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.205 +/- 0.047
Transit depth (Rp/Rs)^2	4.21 +/- 1.92 [%]
Orbital Inclination (inc)	84.37 +/- 1.67
Airmass coefficient 1 (a1)	0.886 +/- 0.019
Airmass coefficient 2 (a2)	0.1 +/- 0.13
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#1 - [371, 179]
Optimal Method	PSF photometry
Transit Duration (day)	0.0421 +/- 0.0094

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.205 ± 0.047 vs 0.1489 (deviation 1.01σ)
Duration fit vs published	0.0421 d vs 0.0779 d (deviation 3.23σ)
Mid-time O–C	15.9 min
Depth SNR	3.7
Residual scatter	2.09 %

Light curve

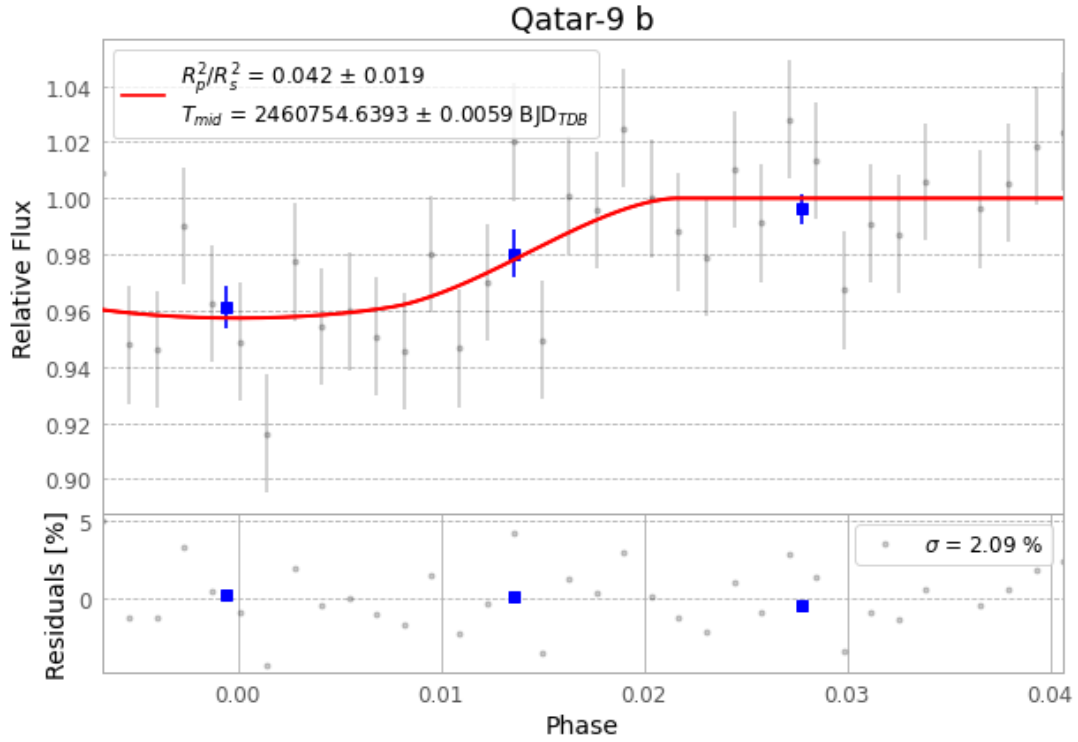


Figure 1 — FinalLightCurve_Qatar-9 b_2025-03-19.png (SHA-256 9a8a8a7f16498974...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6289b41d01928049...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

36 FITS frames (24 MB), Qatar-9250320030015.FITS → Qatar-9250320044515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250320.log (e6e6831671c4...), FinalLightCurve_Qatar-9 b_2025-03-19.png (9a8a8a7f1649...), FinalLightCurve_Qatar-9 b_2025-03-19.csv (88d8977936c9...)

Compute resources

Wall time 71.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:34Z.